

XR Renovation Previewer

Low-fidelity prototype, test plan and studio findings

Kaike Nehme · 7 August 2026 · Digital Prototyping and Extended Reality

01 THE CONCEPT

Phone apps for previewing a renovation — Dulux Visualizer, Houzz, IKEA Kreativ — all fail the same way. They show a candidate colour as a flat rectangle on a six-inch screen, lit by a generic renderer, detached from the room it is meant to go in. The three things that actually decide whether a renovation works are all spatial: **scale**, **adjacency** to what you are keeping, and **light**.

*This project is a phone paint-visualiser app, **but using XR** so you can hold a sample up against the surfaces you are **keeping**, at full size, in your own light. You pull samples directly off the things staying in the room — the floor, the sofa — and those samples generate what works with them. Nothing is picked from a catalogue: the room you already have decides what fits.*

THE ONE PART BEING PROTOTYPED TODAY

Pull a sample off something you're keeping → **hold it up against something you're changing** → **rotate to vary it** → **release to commit**

Not tested today: scheme comparison · marking keep/change · MR passthrough · colour-harmony maths · anything involving the headset.

02 THE PHYSICAL PROTOTYPE

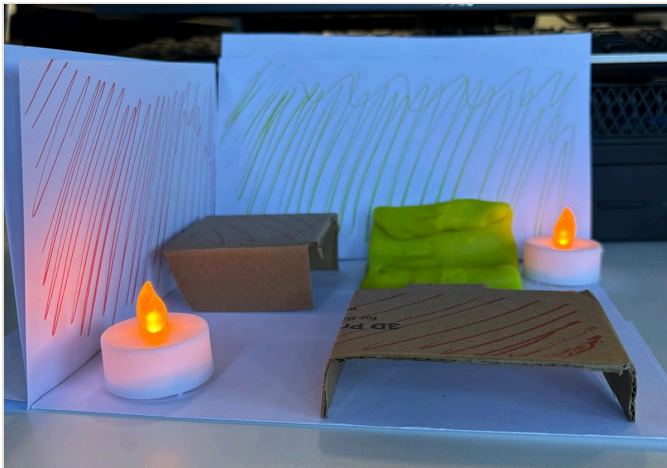


Plate 1 — front. Two hinged paper walls, one scribbled red and one green. Card and playdoh furniture on a paper floor.

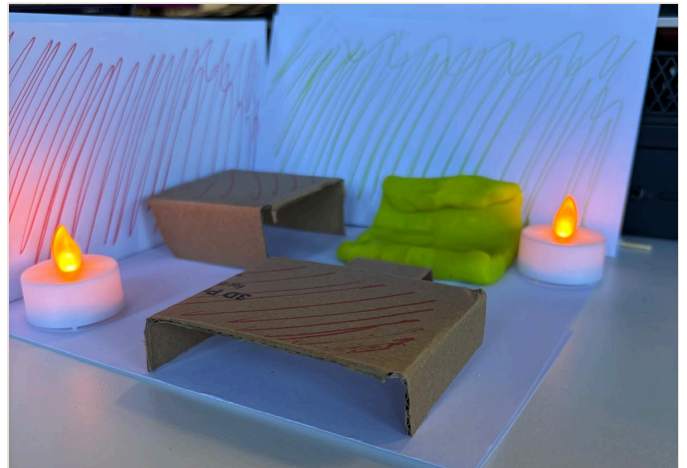


Plate 2 — eye level. Viewed from inside the room, at the height a seated occupant would judge the space from.

What it is made of

 Red wall	CHANGEABLE	Paper panel, hand-scribbled. Lifts off and is swapped by the wizard to simulate a colour change.
 Green wall	CHANGEABLE	Second panel at right angles, so two candidate colours can be judged in one glance.
 Card armchair	KEPT	Left as bare kraft card — uncoloured, so it reads as the existing furniture that is staying. The sample source.
 Playdoh sofa	CHANGEABLE	Moulded by hand. Colour deliberately matched to the green wall — tests whether the loop reads the same on furniture as on walls.
 Card table	CHANGEABLE	Corrugated card, scribbled red to match the red wall. The second half of the coordination test.
 LED tea lights x2	LIGHT	Warm point sources at floor level, against cool ambient room light.

What emerged from building it

LIGHT	The tea lights were not planned, and they are the most valuable thing here. In both plates the red wall reads warm orange near the flame and cool pink further away — the same paper, two apparent colours. This is precisely the affordance the concept claims XR has over a phone, demonstrated in cardboard before a line of code exists.
MATCHING	Coordination happened by hand, unprompted. The sofa was matched to the green wall and the table to the red one while building. The instinct the concept is designed around turned up on its own, which is weak but real evidence that the premise is sound.
MATERIALS	Three materials, three jobs. Paper = the changeable surface. Card = the fixed structure. Playdoh = form that can be reshaped as well as recoloured, which raises a question the digital version has not answered: should furniture be <i>re-shapeable</i> , or only recolourable?

ADD BEFORE TESTING — the card armchair must be visibly labelled as **staying**. If nothing marks it, participants cannot discover a rule the prototype does not express, and question Q1 below returns a false negative. A strip of tape or a written label is enough.

03 THE INTERACTION BEING TESTED

STEP 01 Pull Touch a surface you are keeping — the bare card armchair — and draw your hand away. A sample of its colour comes with you.	STEP 02 Hold up Raise the sample toward a surface you want to change. As it nears, that surface previews the colour behind it.	STEP 03 Tune Rotate the sample in your hand to move through the range that works with the thing you pulled it from. Warmer, cooler, deeper.	STEP 04 Commit Release. The preview becomes the surface. Move away without releasing and it reverts.
---	--	---	--

There are no menus anywhere in that loop. Every change is a physical act on the thing being changed — a direct response to the brief's warning that XR interactions must not be "limited to simply pushing VR buttons".

04 WHAT WE NEED TO FIND OUT

QUESTION	IF THE ANSWER IS NO
Q1 Do people work out that samples come from the things being kept ?	Kept surfaces need a visual affordance
Q2 Is holding a sample near a surface understood as "preview"?	Proximity-preview fails; needs an explicit trigger
Q3 Do people rotate the sample to ask for a variation?	Twist-to-tune is arbitrary — cut it or redesign it
Q4 Does the same gesture read correctly for furniture as for walls?	The loop does not generalise; scope is at risk
Q5 Do fewer, constrained options feel helpful or restrictive?	The core premise is wrong

Assumptions under challenge

A1 Discoverability. The loop can be performed without instruction.

A2 Constraint as help. Generating options from what is staying *reduces* choice paralysis rather than causing frustration.

Q5 / A2 matters most. It can genuinely fail, and a failure is a finding worth reporting — not a problem to hide.

05 METHODOLOGY

Wizard-of-Oz play-acting on a physical low-fidelity prototype. A second person acts as the system: when the participant performs a gesture, the wizard physically swaps the paper to produce what the app *would* have done. The participant is never told how it works. This tests the interaction model before any code exists — and discoverability cannot be measured if you explain the interface first.

Participant	Wizard	Observer
Uses the prototype. Thinks aloud throughout.	Silently swaps paper per the rules. Never explains.	Times, tallies, records verbatim quotes. Says nothing.

Wizard response rules — follow exactly, every run

PARTICIPANT DOES	WIZARD DOES
Holds a swatch near a surface	Swap that surface's panel to the swatch colour. Hold it there.
Moves the swatch away	Put the original panel back.
Rotates the swatch in hand	Hand over a different swatch from the same harmony set.
Releases it onto a surface	Leave the new panel in place — committed.
Taps or presses a surface	Do nothing. Observer records it.
Touches the kept armchair	Hand them a swatch matching its colour.
Tries to sample a changeable surface	Do nothing. Observer records it.
Asks how it works	"What would you expect?" Nothing more.

06 STEP-BY-STEP TESTING PROCESS

One run ≈ 6 min 30 s. Aim for three to four runs. Reset the room to its starting state between participants and photograph it, so everyone begins identically.

TIME	STEP	SCRIPT AND WHAT IS OBSERVED
0:30	Setup	Room on table, kept item labelled, swatch deck to one side. Reset any previous changes.
0:30	Brief	"This is a living room. The armchair is staying. Everything else can change. Think out loud." Nothing about how it works.
1:30	Free exploration	No task given. Observe first actions, whether they touch the kept item, whether they try to tap. → D1, D2, D4, D6
2:00	Task 1 — colour	"Change the room so it works with the armchair you're keeping." Wizard responds per rules. → D3, D5, D7
1:00	Task 2 — furniture	"Now change a piece of furniture the same way." Tests Q4.
1:00	Post-task	D8, then: "What did you think would happen when you held the card up?" and "Where did you think the colours came from?"

07 DATA COLLECTED AND HOW

MEASURE	TYPE	CAPTURED BY
D1 First three actions attempted, in order	sequence	Observer writes them down
D2 Time to first correct pull from the kept item	seconds	Phone stopwatch
D3 Prompts needed before the loop was performed	tally	Observer
D4 Tried to tap or press instead of holding up?	yes / no	Observer
D5 Rotated the swatch unprompted?	yes / no	Observer
D6 Tried to sample a <i>changeable</i> surface?	yes / no	Observer
D7 Confusion quotes, verbatim	qualitative	Observer, in the participant's own words
D8 "Did fewer options feel helpful or restrictive?" 1–5 + why	mixed	Asked after the task
D9 Photograph of each configuration produced	visual	Phone

D7 and D8 carry the most weight. Numbers from three or four participants prove nothing statistically — the value is entirely in *why* people got stuck.

08 RESULTS — CAPTURE SHEET, CARRIED TO IP1

The Week 2 session produced **peer design feedback rather than protocol data** — the Wizard-of-Oz run was not executed against the timed process above, so no behavioural measures were captured. Q1–Q5 remain open and this sheet carries forward to the Week 5 studio test.

MEASURE	P1	P2	P3	P4
D1 First three actions				
D2 Time to first correct pull				
D3 Prompts needed				
D4 Tapped instead of held up?				
D5 Rotated unprompted?				
D6 Sampled a changeable surface?				
D8 Helpful / restrictive (1–5)				

FINDING	EVIDENCE
Q1 Samples come from kept things	
Q2 Proximity reads as preview	
Q3 Rotation reads as "vary it"	
Q4 Works for furniture too	
Q5 Constraint helps vs restricts	

09 WHAT CHANGED AFTER PEER FEEDBACK

Three requests came out of the studio. All three arrived phrased as "add a button" — two had better answers underneath.

ADOPTED
IP1

Light control, as a diegetic object. Peers asked to switch lights on and off. This was not in the concept — the tea lights were unplanned. The request independently confirms the concept's central claim, arrived at by someone who did not know that was the claim. Built as a lamp you touch or a wall switch, never a floating button, so the no-menus rule holds. Extended to three states — **warm bulb / cool bulb / daylight** — which makes "does this green survive a warm bulb at 9pm?" testable at no extra interaction cost.

ADOPTED
IP2A

Furniture repositioning. Peers asked to rotate furniture more easily. First reading was that this might be an artefact of tabletop scale — you cannot walk around a model on a desk, so spinning it is the only way to see another angle. Checked with the session: the intent was **genuine layout change**. Grab, move and rotate is therefore a real system, scheduled for Week 9 and deliberately excluded from IP1 so it cannot compete with light for build time on a hurdle assessment.

ADOPTED
IP2B

VR↔MR as a runtime toggle. Peers wanted to change between the virtual and real-room environments at will. The plan already moved VR → MR across prototypes, but one-way. Reframed as a live switch: see the scheme in your real room via passthrough, then in an idealised virtual version — honest against aspirational, photograph against magazine. Doubles as the "significantly different interactions" IP2b requires.

Unchanged

The core loop — pull, hold up, tune, commit — drew no criticism. **Constraint-generated options was not tested at all**, and remains the single most important open question; assumption A2 is where IP1's testing time should go.

New questions created by the additions

Q6 Can **tuning rotation** (twisting the held sample) and **layout rotation** (turning a sofa) share a gesture vocabulary without colliding, now that both exist? Two meanings for one wrist movement, separated only by what is in your hand, is exactly the ambiguity a participant reads as a bug.

Q7 Are three lighting states enough, or is continuous time-of-day needed to make the light claim properly testable?

WHY THIS SECTION MATTERS — Criterion 3 of the Design Concept Report rewards showing how the initial idea was **refined**, not how right it was to begin with. A request interrogated and redirected scores better than a feature list accepted wholesale.